

Domino Neuron Relay

Test a domino relay across a gap and name the model's limits

At a glance: Test a domino relay across a gap and name the model's limits. Plan about 60 minutes for the core block; 4 to 6 teams of 4 to 5 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain signal propagation and the model's limits: myelin insulates one axon; synapses connect different cells; systems reliability (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 25 STUDENTS AND 6 TEAMS

- Dominoes
- Rulers
- Tape
- Index cards

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Tape down and stage one shared backup track, then rotate all six teams through it sequentially.
- 02 Stage the scoring sheet and any reference values before testing begins.
- 03 Stage only the activity-specific safety controls named in the student safety note and staff run sheet; gloves, goggles, and trays are not universal requirements.
- 04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Test a domino relay across a gap and name the model's limits." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:38

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record

Teams run the standard test and log a baseline result before changing anything.

Phase 6 Redesign

Each team changes exactly one variable, predicts the effect, and re-tests to compare against baseline.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. PPE off, wash or wipe down.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: No measurable result on the first test. **Fix:** Check the setup against the steps, confirm the standard test was followed, and log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to signals travel in a chain.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- What happened when the gap got too wide?
- Why is the bridge only an engineering repair rather than a literal synapse or myelin sheath?
- Why does myelin speed real signals?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Transmission success	35
Gap challenge	25
Explanation	20
Redesign	20
Total	100

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES Exploratorium Domino Effect